Screen Time, Sleep, and Cognition

A buffering role for sleep in toddlers' cognitive development

INTRODUCTION

- **Estimates of toddlers' screen time** have surged beyond recommendations of 1 hour per day.
  Egan & Murray, 2014; McArthur et al., 2022
- **Executive functions** are cognitive skills that allow control of thoughts, behaviour, and emotions.
  Stucke & Doebel, 2024; Zelazo & Carlson, 2012
- **Given the highly-stimulating nature of digital media**, there are concerns about its potential impact on toddlers' cognitive development.
  Coyne et al., 2021; Lillard & Peterson, 2011

The aim of this study was to explore the role of sleep within the association of toddlers' screen time and executive functions.

METHODS

Data for this study was collected via an online survey widely advertised to parents and caregivers of toddlers (18 to 36 months) across the island of Ireland from February 21st to March 25th, 2025.

Sample

- 192 Toddlers (53% female)
- *M* = 25.2 months (*SD* = 5.33)
- 95% White, 5% Non-White

Respondents

- 99% Mothers
- 93% Third-Level Education

Respondents were asked to complete an online survey on their toddler's executive functioning, screen time & sleep.

Early Exec. Functioning Questionnaire
(Hendry & Holmboe, 2020)

A parent-reported measure of executive function skills.

- **Cognitive Executive Functioning**: Executive skills used in situations with less emotional weight (e.g., reasoning).
- **Executive Function Regulation**: Executive skills used in emotionally charged situations (e.g., calming down).

Brief Infant Sleep Questionnaire
(Mindell et al., 2019)

A parent-reported measure of sleep quality in infants and toddlers.

- **Sleep Quality**: The ease with which toddlers fall asleep, the duration & interruptions of sleep.

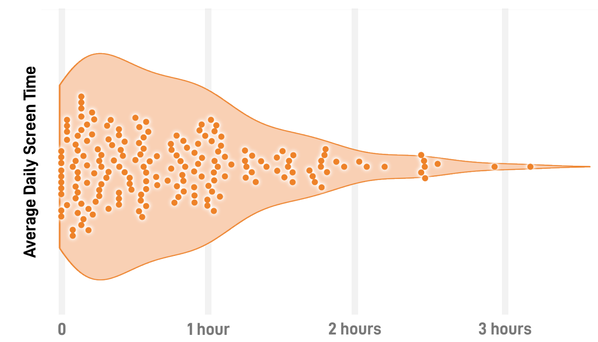
SCREENS - Questionnaire
(Klakk et al., 2020)

A parent reported measure of toddlers' average, daily screen-time.

RESULTS

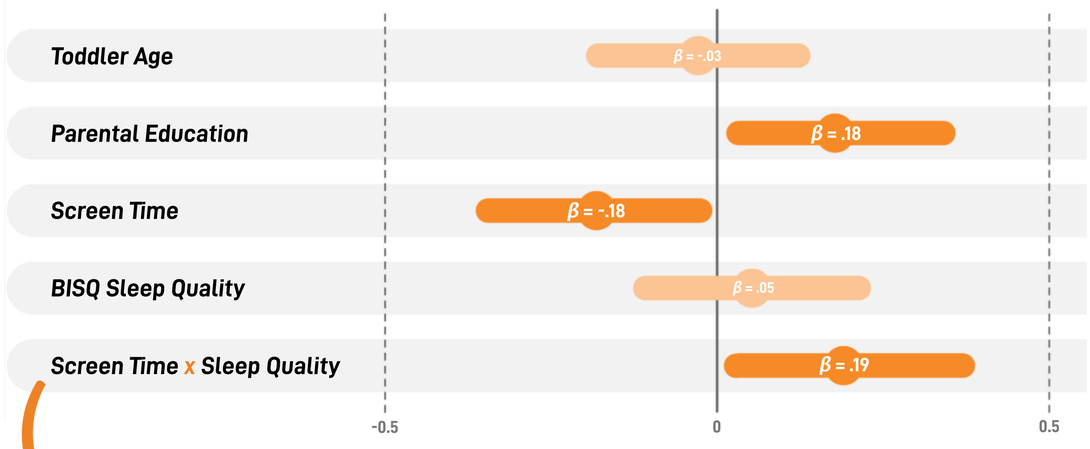
1 **Average Screen Time**
47.3 min/day

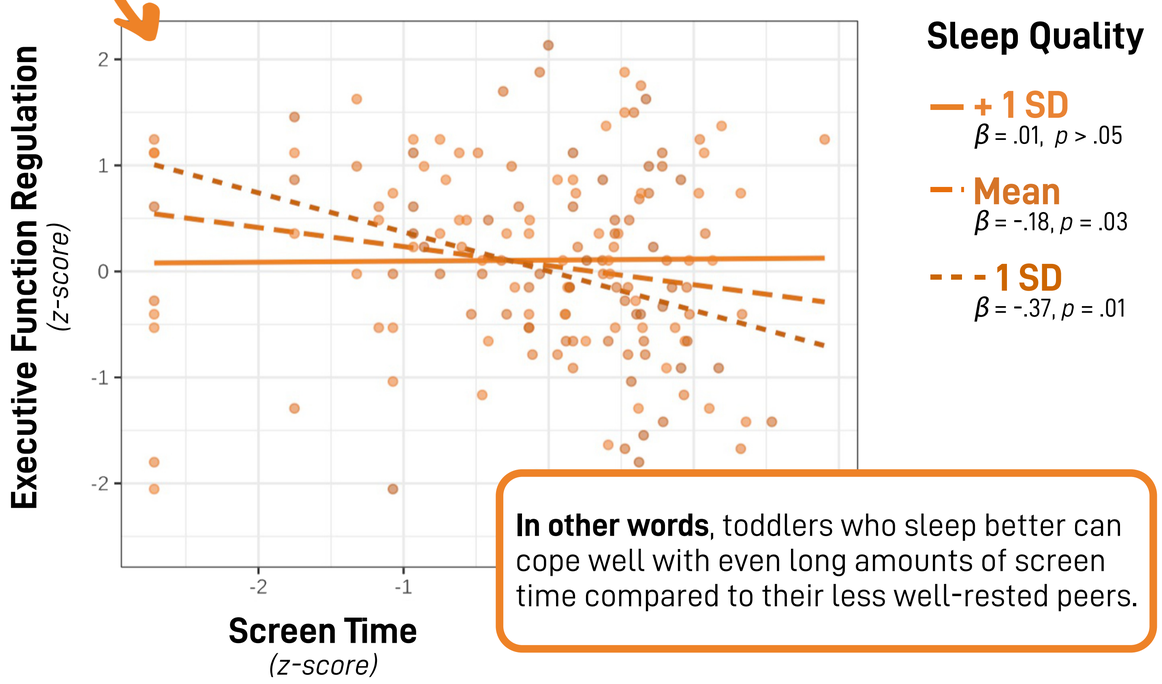
Toddlers' daily screen time ranged from zero to three hours per day.



2 Results from a multiple regression analysis demonstrated a **significant interaction** between screen time and sleep quality on **toddlers' executive function regulation**.

Multiple regression model predicting **Executive Functioning Regulation.**
F(5, 151) = 2.63, *p* = .03 | R^2 = .08, Adj. R^2 = .05





In other words, toddlers who sleep better can cope well with even long amounts of screen time compared to their less well-rested peers.

3 **Cognitive executive functioning** was not affected by screen time after sleep quality was taken into account.

CONCLUSION

- **Digital media is popular even with young children.** Though the average screen time in this study is below the recommended threshold, the data shows that there are a range of attractive digital activities for toddlers.
- **Sleep buffers the negative role of toddlers' screen time** on their executive function regulation development. Consequently, sleep is an important variable to consider within further research and practice.

This study shows that contextual factors of toddlers' screen time, such as the quality of their sleep, need to be considered when assessing the role of screen time in children's development.

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The authors have **no conflict of interest** to declare.